Reference investigation

Observations

- Corpus: corpus_capture_01.pcapng (capture_id sha256:1a2f3637124abcd43049bdf9432ae3d18024276cf7de29efd43562e0c61f272c).
- Rule v1 scope: A_to_B.
- Messages framed: 280.
- Counts: matched=60, mismatched=80, incomplete=0, ambiguous=0, uncovered=0, not_applicable=140, outdated=0, unknown=0
- First bytes of the primary capture, as the observation the layout was read from:

```
s1 A_to_B: 0100000110010000011101000001120100000110010000001
```

Framing

The length field is at offset 2, two bytes, big-endian. What it counts is not written down, so the three candidates are framed against the real bytes. A candidate is kept only if it frames every stream with no leftover bytes, no truncated final message and no impossible command byte.

capture	stream	bytes	payload	payload_and_length_field	entire_message
corpus_capture_01.pcapng	s1 A_to_B	300	60 msgs, 300/300 B, clean	1 msgs, 0/300 B, broken	1 msgs, 0/300 B, broken
corpus_capture_01.pcapng	s1 B_to_A	420	60 msgs, 420/420 B, clean	3 msgs, 420/420 B, broken	1 msgs, 0/420 B, broken
corpus_capture_01.pcapng	s2 A_to_B	420	60 msgs, 420/420 B, clean	2 msgs, 420/420 B, broken	1 msgs, 0/420 B, broken
corpus_capture_01.pcapng	s2 B_to_A	420	60 msgs, 420/420 B, clean	2 msgs, 420/420 B, broken	1 msgs, 0/420 B, broken
corpus_capture_01.pcapng	s3 A_to_B	100	20 msgs, 100/100 B, clean	1 msgs, 0/100 B, broken	1 msgs, 0/100 B, broken
corpus_capture_01.pcapng	s3 B_to_A	1376	20 msgs, 1376/1376 B, clean	3 msgs, 1376/1376 B, broken	2 msgs, 1376/1376 B, broken
corpus_capture_02.pcapng	s1 A_to_B	100	20 msgs, 100/100 B, clean	1 msgs, 0/100 B, broken	1 msgs, 0/100 B, broken
corpus_capture_02.pcapng	s1 B_to_A	140	20 msgs, 140/140 B, clean	2 msgs, 140/140 B, broken	1 msgs, 0/140 B, broken
corpus_capture_02.pcapng	s2 A_to_B	140	20 msgs, 140/140 B, clean	2 msgs, 140/140 B, broken	1 msgs, 0/140 B, broken
corpus_capture_02.pcapng	s2 B_to_A	140	20 msgs, 140/140 B, clean	2 msgs, 140/140 B, broken	1 msgs, 0/140 B, broken

Only `payload` frames all 10 streams with no leftover bytes and no truncated message. The other two candidates either stop after the first size or split the stream into a handful of oversized blocks, so they are rejected.

Counterexamples

80 counterexamples against rule v1:

- status=mismatched session=s2 direction=A_to_B message_offset=0 field=command reason=value 2 not in expected [1] bytes_hex=02000003130055
- status=mismatched session=s2 direction=A_to_B message_offset=7 field=command reason=value 2 not in expected [1] bytes_hex=020000031400ff
- status=mismatched session=s2 direction=A_to_B message_offset=14 field=command reason=value 2 not in expected [1] bytes_hex=0200000315009f
- status=mismatched session=s2 direction=A_to_B message_offset=21 field=command reason=value 2 not in expected [1] bytes_hex=0200000313005c

- status=mismatched session=s2 direction=A_to_B message_offset=28 field=command reason=value 2 not in expected [1] bytes_hex=02000003140003
- status=mismatched session=s2 direction=A_to_B message_offset=35 field=command reason=value 2 not in expected [1] bytes_hex=0200000315002b
- status=mismatched session=s2 direction=A_to_B message_offset=42 field=command reason=value 2 not in expected [1] bytes_hex=02000003130021
- status=mismatched session=s2 direction=A_to_B message_offset=49 field=command reason=value 2 not in expected [1] bytes_hex=02000003140010
- status=mismatched session=s2 direction=A_to_B message_offset=56 field=command reason=value 2 not in expected [1] bytes_hex=020000031500b1
- status=mismatched session=s2 direction=A_to_B message_offset=63 field=command reason=value 2 not in expected [1] bytes_hex=020000031300e2
- status=mismatched session=s2 direction=A_to_B message_offset=70 field=command reason=value 2 not in expected [1] bytes_hex=0200000314003c
- status=mismatched session=s2 direction=A_to_B message_offset=77 field=command reason=value 2 not in expected [1] bytes_hex=020000031500eb
- ... and 68 more of the same kind (each one is a command byte outside the expected set).

Refinement

Rule refined to v2. Changes:

- field command: expected [1] -> [1, 2, 3]

Counts after refinement: matched=140, mismatched=0, incomplete=0, ambiguous=0, uncovered=0, not_applicable=140, outdated=0, unknown=0.

Version differentiation

Rule v1 and rule v2 are different rules over the same bytes: nothing about the capture changed, only the description of it. The counts below come from re-running both versions.

metric	v1	v2	delta
matched	60	140	+80
mismatched	80	0	-80
counterexamples	80	0	-80
coverage	0.5	0.5	+0.0
precision	0.428571	1.0	+0.571429
counterexample_density	57.142857	0.0	-57.142857

Resolved counterexamples: 80. Introduced counterexamples: 0. Coverage delta: +0.0.

A rule change never rewrites the bytes it was checked against. The v1 result stays on disk with its own rule version; a reader sees which version produced which verdict.

Refined rule

The refined rule is written to `corpus_rule_v2.json` so it can be applied again by the CLI without re-deriving it.

Applicability

Rule v2 applied to corpus_capture_02.pcapng (capture_id sha256:4aa281b97de9afa09d74ff0f5c41e9f6c4fc2b7b07ddf314f33b55d2fec95afc). Counts: matched=40, mismatched=0, incomplete=0, ambiguous=0, uncovered=0, not_applicable=40, outdated=0, unknown=0. Counterexamples: 0.

Applicability boundary

Rule v2 applied to synthetic_live.pcapng, a capture produced by a real TCP stack with a deliberately different layout (little-endian length, a transaction id byte, commands 0x21/0x22/0x23). The rule was never fitted to this traffic.

- s1 A_to_B: 61 B -> 1 messages, 61/61 B, truncated
- s1 B_to_A: 116 B -> 1 messages, 116/116 B, truncated

Streams framed cleanly: 0/2. Counts: matched=0, mismatched=0, incomplete=1, ambiguous=0, uncovered=0, not_applicable=1, outdated=0, unknown=0 Counterexamples: 1.

The big-endian length and the different command set mean the rule does not transfer: its matches here are coincidental, not evidence. This is the boundary of the rule's applicability, and it is stated rather than hidden by re-tuning until something matches.

Journal correlation

Journal: 185 entries; the timestamp is the capture time of the request packet. Every framed request is matched to the nearest entry inside a 500 ms window.

capture	direction	requests	correlated	unmatched requests	unmatched entries	target=parameter	value=result
corpus_capture_01.pcapng	A_to_B	140	140	0	45	140	0/0
corpus_capture_02.pcapng	A_to_B	40	40	0	145	40	0/0

Every request lands on a journal entry at the same timestamp, and the `target` field resolves to the parameter the journal names for every correlated request. The `value` field is present only where the request carries a written value; the expected `result` holds only for unmatched entries, so a differing value is kept as evidence against the field, not hidden.

Alternative readings

Fields marked `hypothesis: true` are named on a guess. Each is scored against competing readings over the framed corpus; the declared meaning is one candidate and may lose. Score is support / (support + contradict).

Field `command` (140 values):

reading	support	contradict	score
entropy_enum	140	0	1.00
low_cardinality	140	0	1.00
offset_shift_+1	140	0	1.00
counter	137	2	0.99
constant	60	80	0.43
message_length	0	140	0.00
remaining_length	0	140	0.00

Best fit: `entropy_enum`.

Field `target` (140 values):

reading	support	contradict	score
low_cardinality	140	0	1.00
constant	20	120	0.14

Best fit: `low_cardinality`.

Field `value` (60 values):

reading	support	contradict	score
entropy_parameter	60	0	1.00
constant	1	59	0.02
message_length	1	59	0.02
remaining_length	0	60	0.00

Best fit: `entropy_parameter`.

Open questions

- Field meanings marked `hypothesis: true` remain assumptions; a rule that matches is not proof.
- Behaviour outside the tested captures is unknown; the rule is only confirmed within the tested domain.

Limitations

- The rule describes the request direction only; responses are framed but not interpreted field by field.
- Field meanings marked `hypothesis: true` (the enum `target`, the payload `value`) remain assumptions. Matching is not proof.
- The framing question was settled against the corpus; a stream whose length field counts something else would need a new rule.
- Ambiguity and gap diagnostics are surfaced, never zero-filled, so a message over a gap is reported `incomplete` rather than guessed.
- The synthetic live capture shares no byte layout with the corpus, so nothing here is claimed to generalise to it.